Input Form (IF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: *LTZGLUE* | ***Context:*** *Luxembourg Public Administration — Civil Servant NLU Deployment*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Form mismatch is moderate. Both benchmark and deployment are text-only in Latin script, so there is no modality or script-direction violation [regional context infrastructure_notes.deployment_modality; writing_systems]. Tokenization (BPE, vocab 50,368, max sequence 1,024) [Q128, Q129] is suitable for typical citizen-correspondence lengths (40–400 words covers many but not all messages [regional context infrastructure_notes.expected_input_length, Q47]). However, signal-distribution mismatch persists at the form level: task inputs are filtered to remove non-Luxembourgish text via OpenLID and constrained to 40–400 words [Q47], which removes precisely the multilingual and length-extreme inputs the deployment will encounter. The benchmark's text register (formal news, dictionary examples, transcribed political speeches) [Q74, Q16, Q30] differs from informal, orthographically variable citizen messages. IF is MODERATE priority; the form-level alignment is partial because text-text matching is achieved, but distributional filtering and register narrowness reduce external validity.

ID	Evidence
Q16	“To construct this dataset, we use RTL news articles. We keep only documents from the twenty most frequent categories. We then filter articles by body length and title length, remove exact duplicate titles, randomly shuffle the remaining instances, and retain a fixed subset of 30k examples.” (p.3)
Q30	“The sentences are derived from the Luxembourgish Online Dictionary (LOD) and are manipulated using the tags available in the dataset.” (p.3)
Q47	“We applied a series of preprocessing steps to ensure data quality. Specifically, we removed articles identified as non-Luxembourgish by OpenLID (Burchell et al., 2023), as well as those containing fewer than 40 words or more than 400 words.” (p.4)
Q74	“The pre-training set is compiled from a variety of sources of LTZ. A large portion of the data stems from RTL (see Section 3), including news articles (News), transcribed radio interviews (Radio), and user comments (Comments). We also include transcribed podcasts (Podcasts) and transcribed political speeches and debates from the Chambre des Députés (Chamber). In addition, we use 1M sentences from the web crawl of the Leipzig Collection (Web, this excludes RTL), text crawled from LTZ chat rooms (Webchat), a Wikipedia crawl from October 2023 (Wikipedia), and finally, example sentences from the LOD retrieved in March 2024.” (p.6)
Q128	“Both models share a GPTNeoXTokenizerFast tokenizer (Black et al., 2022), a BPE-based tokenizer, which we train on the entire pre-training set, using a minimum frequency of two and a vocabulary size of 50,368.” (p.12)
Q129	“We use a constant batch size of 1024 packed sequences, where both models have a max sequence length of 1024.” (p.12)

Strengths

- Text-only modality matches deployment exactly [regional context infrastructure_notes.deployment_modality]
- Latin-script writing system aligns with all three administrative languages [regional context writing_systems]
- BPE tokenizer trained on broad LTZ pre-training corpus including comments, webchat, podcasts may handle orthographic variation reasonably [Q74, Q128]
- Sequence length cap of 1,024 tokens is sufficient for most short-to-medium correspondence

ID	Evidence
Q74	“The pre-training set is compiled from a variety of sources of LTZ. A large portion of the data stems from RTL (see Section 3), including news articles (News), transcribed radio interviews (Radio), and user comments (Comments). We also include transcribed podcasts (Podcasts) and transcribed political speeches and debates from the Chambre des Députés (Chamber). In addition, we use 1M sentences from the web crawl of the Leipzig Collection (Web, this excludes RTL), text crawled from LTZ chat rooms (Webchat), a Wikipedia crawl from October 2023 (Wikipedia), and finally, example sentences from the LOD retrieved in March 2024.” (p.6)

Q128	“Both models share a GPTNeoXTokenizerFast tokenizer (Black et al., 2022), a BPE-based tokenizer, which we train on the entire pre-training set, using a minimum frequency of two and a vocabulary size of 50,368.” (p.12)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Both source and target are text. Length filter (40–400 words) [Q47] excludes very short and very long messages that are realistic in administrative correspondence (e.g., one-line inquiries, multi-page complaints).

IF-2: Yes — Luxembourg has 99.0% internet penetration [WEB-7] and advanced digital infrastructure [regional context infrastructure_notes.connectivity]; data-capture specifications match.

IF-3: Use-case-specific differences: filtering of non-Luxembourgish text via OpenLID [Q47] removes code-switched inputs that the deployment must actually process. Pre-training data is broader [Q74] but downstream task data is narrower in register.

IF-4: External-validity concerns from (a) length filtering excluding tail of distribution, (b) language-detection filtering removing multilingual inputs, (c) register narrowness in downstream task data [Q47, Q112].

ID	Evidence
Q47	“We applied a series of preprocessing steps to ensure data quality. Specifically, we removed articles identified as non-Luxembourgish by OpenLID (Burchell et al., 2023), as well as those containing fewer than 40 words or more than 400 words.” (p.4)
Q74	“The pre-training set is compiled from a variety of sources of LTZ. A large portion of the data stems from RTL (see Section 3), including news articles (News), transcribed radio interviews (Radio), and user comments (Comments). We also include transcribed podcasts (Podcasts) and transcribed political speeches and debates from the Chambre des Députés (Chamber). In addition, we use 1M sentences from the web crawl of the Leipzig Collection (Web, this excludes RTL), text crawled from LTZ chat rooms (Webchat), a Wikipedia crawl from October 2023 (Wikipedia), and finally, example sentences from the LOD retrieved in March 2024.” (p.6)
Q112	“Coverage across domains, registers, and demographic varieties may also be limited. LTZ displays substantial orthographic and sociolinguistic variation, yet most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts.” (p.9)
Q129	“We use a constant batch size of 1024 packed sequences, where both models have a max sequence length of 1024.” (p.12)
WEB-7	99.0% internet penetration — confirms text-output channel is universally accessible (https://datareportal.com/reports/digital-2024-luxembourg)

Information Gaps

- Empirical length distribution of expected citizen correspondence
- Empirical proportion of code-switched messages in real digital channels

Requires Expert Verification

- Sample-level review of incoming citizen messages to validate that benchmark form constraints (length, language-purity filtering) do not exclude operationally important inputs

---

Remediation

Gap 1: Length filter (40–400 words) and language-purity filter (OpenLID) exclude tail-distribution inputs the deployment will face

Recommendation: Construct a supplementary evaluation slice without length and language-purity filters to characterize model behavior on very short, very long, and multilingual inputs; report performance separately on these slices.